

# Supplementary Materials for Prior-guided image analysis for Bayesian longitudinal maize height phenotyping with fixed cameras

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This supplement gives the detailed implementation, causal information set, later-year camera-pair mapping, calibration decisions, uncertainty construction, candidate audit, and reproducibility provenance. Author identifiers are confined to the separate title page.

## **S1. Later-year 2024 manual-height validation**

The source workbook was imported without numerical imputation or correction. It contains 247 nonmissing manual field heights from 45 plants in four biological rows. The Camera layout worksheet establishes that each biological row is covered by a left/right camera pair. Right-view local plant identities 1–6 map to row-wide right-to-left identities 1–6, and left-view identities map to 7–12. Two camera halves did not yield usable automated output. After same-date quality control, 166 records from 34 plants, four biological rows, and six camera views remained.

The two views are not treated as independent rows. All bootstrap replicates sample the biological row and retain both camera views, every plant, and all dates inside that row. The complete camera accounting is in `outputs/manual_height_transfer_2024/camera_pair_accounting_2024.csv`; the imported layout is in `data/raw/manual_height_2024/camera_layout_2024.csv`.

The analysis code was locked at commit `59b42b74596c598017fdb0cd4b80af1aec7649d6` before the workbook was opened for scoring. The transfer equation was fitted on 61 recovered-source 2021 records from six camera rows. The recorded camera-distance ratio, 8.5/10.25, maps the 2021 scale to 2024. The 2021 leave-one-camera-row-out RMSE, 15.58 cm, is retained as a nonshrinking transfer component. Student-*t* intervals use five degrees of freedom. Every 2024 camera is processed in time order; manual outcomes are joined only after single-frame and Bayesian predictions are emitted.

The Bayesian and Gaussian filters use the same  $N(3, 2^2)$  cm/day initial growth distribution, 8.78 cm/ $\sqrt{\text{day}}$  height-process scale, and 0.35 cm/day/ $\sqrt{\text{day}}$  growth-process scale. The Gaussian observation variance is the squared 2021 transfer RMSE. Other fixed comparators are EWMA ( $\alpha = 0.5$ ), a three-image running median, and Holt level-trend ( $\alpha = 0.5$ ,  $\beta = 0.2$ ). All comparators process the complete daily prefix before manual outcomes are joined.

Table S1: Contribution of each 2024 biological row to the later-year manual validation. Positive MAE gain favors Bayesian longitudinal estimation.

| Biological row | Records | Plants | Single-frame MAE (cm) | Bayesian MAE (cm) |
|----------------|---------|--------|-----------------------|-------------------|
| M-0006         | 48      | 11     | 19.87                 | 18.38             |
| M-0013         | 66      | 12     | 12.74                 | 11.61             |
| W-0010         | 25      | 5      | 28.98                 | 28.74             |
| W-0015         | 27      | 6      | 18.00                 | 19.08             |

Overall single-frame MAE was 18.10 cm and Bayesian MAE was 17.36 cm; the corresponding RMSEs were 23.54 and 22.41 cm. Correlation increased from 0.813 to 0.847. The 0.74-cm paired MAE gain had a four-row bootstrap interval of -0.44 to 1.30 cm. Transfer-aware coverage at 50%, 80%, 90%, and 95% was 53.0%, 78.9%, 90.4%, and 97.0%, with mean widths 27.6, 56.0, 76.4, and 97.5 cm.

Table S2: Fixed causal comparator results against 2024 manual field height. Baseline-minus-Bayesian MAE is positive when the Bayesian estimate has lower MAE.

| Estimator             | Bias (cm) | MAE (cm) | RMSE (cm) | Baseline minus Bayesian MAE (cm) |
|-----------------------|-----------|----------|-----------|----------------------------------|
| Single frame          | -13.49    | 18.10    | 23.54     | 0.74                             |
| EWMA                  | -16.46    | 19.70    | 24.44     | 2.34                             |
| Running median        | -16.26    | 19.94    | 24.66     | 2.58                             |
| Holt level-trend      | -13.65    | 17.60    | 22.63     | 0.24                             |
| Gaussian local-linear | -13.22    | 17.35    | 22.42     | -0.01                            |
| Bayesian longitudinal | -13.24    | 17.36    | 22.41     | 0.00                             |

The manual and image acquisitions were matched by calendar date. Their absolute time difference had median 2.78 h and maximum 6.90 h. The source protocol provides two possible pre-tasseling definitions but the workbook does not identify which was used. The outcome is therefore labeled manual field height rather than a particular anatomical landmark.

An associated metadata audit found that 89 of 237 accepted red-component candidate fits implied more than the documented maximum of 10 one-foot pole intervals. The 2024

manual-height comparison consequently excludes that automatic enumeration and uses the 2021 transfer plus recorded camera geometry. Files sufficient to reproduce this decision and the validation are under `outputs/manual_height_transfer_2024/`.

## S2. Detailed implementation specification

### *S2.1. What is and is not Bayesian in the field implementation*

Two sequential operations must be distinguished. The field association code maintains a running horizontal position and base row for each plant. It uses a fixed update gain of 0.5 and does not propagate a position covariance, calculate a data-dependent Kalman gain, or retain position particles. This is a fixed-gain state update, algebraically equivalent to a steady-state Kalman-mean recursion after the gain has been specified. It is not a full Bayesian posterior over position.

The height and growth component is Bayesian. A weighted particle approximation represents the joint distribution of latent height and growth rate. Particles are propagated through the transition model, weighted with the inlier/outlier measurement likelihood, normalized, and resampled when the effective sample size crosses its prespecified threshold. Posterior summaries and one-step predictive distributions are calculated from those weighted particles.

The term *prior-guided image analysis* refers to feedback across the image/model boundary, not simply to the presence of priors in a Bayesian model. The stronger mechanism in which the predicted height distribution re-ranks competing top-keypoint candidates is evaluated in the controlled candidate-ambiguity simulation and in a matched real-image audit. It was not the top-keypoint rule used to produce the primary field-height results. Thus the field implementation supports claims about prior-guided plant association and base-keypoint updating, while the simulation establishes that a height prior can improve top-candidate choice under known ambiguity. The real-image audit tests whether the available natural candidate sets actually contain that ambiguity.

Table S3: Full parameter and implementation specification for the two empirical analysis streams. A dash indicates that the operation was not applied in that stream.

| Component           | Quantity or operation                   | 2024–2025 daily<br>pole stream                                                                   | 2021<br>development-era<br>manual-reference<br>analysis                                                                  |
|---------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Observation<br>unit | Plant identity and<br>chronological key | <code>camera_</code><br><code>genotype</code> ,<br>left-to-right plant<br>track, capture<br>time | <code>rowid</code> plus<br><code>plantid</code> , field<br>date                                                          |
| Observation<br>unit | Measurement entering<br>height filter   | Depth-corrected<br>red-band pole<br>calibration, cm                                              | Archived<br>single-image mask<br>extent<br><code>height_sam</code> ,<br>multiplied by the<br>recorded<br>geometric scale |
| Reference           | In-scene physical<br>definition         | Red-band pole;<br>adjacent band<br>edges 1 ft (30.48<br>cm) apart                                | Camera-support<br>pole annotation is<br>an installation<br>asset, not the<br>source of<br><code>height_sam</code>        |

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*Table S3 continued*

| Component              | Quantity or operation | 2024–2025 daily<br>pole stream                                                                                       | 2021<br>development-era<br>manual-reference<br>analysis         |
|------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Reference              | Depth factor          | $D_{\text{tgt}}/D_{\text{ref}} =$<br>$8.5/10.0 = 0.85$                                                               | Separate 2021<br>scale, 0.415886<br>cm/pixel, zero<br>intercept |
| Calibration<br>history | Reference selection   | Best passing<br>calibration with<br>timestamp no<br>later than the<br>current image; fit<br>RMSE capped at<br>0.5 in | Not applicable to<br>archived<br><code>height_sam</code>        |
| Field<br>association   | Expected plants       | 6 per initialized<br>camera view                                                                                     | Archived<br>identities supplied                                 |
| Field<br>association   | Initialization        | First complete<br>six-detection<br>frame, sorted left<br>to right                                                    | –                                                               |

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*Table S3 continued*

| Component            | Quantity or operation | 2024–2025 daily<br>pole stream                                                                 | 2021<br>development-era<br>manual-reference<br>analysis |
|----------------------|-----------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Field<br>association | Assignment            | Minimum total<br>absolute<br>horizontal<br>distance;<br>maximum<br>normalized<br>distance 0.16 | –                                                       |
| Field<br>association | Position/base update  | $m_t =$<br>$(1 - K)m_{t-1} + Kx_t$ ,<br>fixed $K = 0.5$ ; no<br>covariance state               | –                                                       |
| Field<br>association | Empirical diagnostic  | 2018 matches, 48<br>more than frozen<br>first-frame<br>anchors                                 | –                                                       |
| Particle filter      | Latent state          | Height $h_t$ and<br>growth rate $r_t$                                                          | Same                                                    |
| Particle filter      | Transition order      | Propagate $h_t$<br>using $r_{t-1}$ , then<br>update $r_t$                                      | Same                                                    |
| Particle filter      | Number of particles   | 5000                                                                                           | 5000                                                    |

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*Table S3 continued*

| Component       | Quantity or operation | 2024–2025 daily<br>pole stream                                                                       | 2021<br>development-era<br>manual-reference<br>analysis                       |
|-----------------|-----------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Particle filter | Base random seed      | 20260723;<br>deterministic<br>plant-specific seed<br>from SHA-256 of<br>plant label and<br>base seed | 20260724; same<br>deterministic<br>derivation                                 |
| Particle filter | Initial height        | $N(Y_1, \sigma_1^2)$ , clipped<br>to $[0, 450]$ cm                                                   | Same                                                                          |
| Particle filter | Initial growth        | $N(3, 2^2)$ cm/day,<br>clipped                                                                       | $N(0, 8^2)$ cm/day,<br>clipped                                                |
| Particle filter | Height limits         | 0–450 cm                                                                                             | 0–450 cm                                                                      |
| Particle filter | Growth limits         | -4–15 cm/day                                                                                         | -4–15 cm/day                                                                  |
| Particle filter | Height-process SD     | 16.866 cm/ $\sqrt{\text{day}}$ ,<br>estimated from<br>depth-corrected<br>2024 daily<br>increments    | 8.78 cm/ $\sqrt{\text{day}}$ ,<br>robust scale of<br>2021 image<br>increments |
| Particle filter | Growth-process SD     | 0.35<br>cm/day/ $\sqrt{\text{day}}$                                                                  | 0.35<br>cm/day/ $\sqrt{\text{day}}$                                           |

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*Table S3 continued*

| Component       | Quantity or operation     | 2024–2025 daily<br>pole stream                                                         | 2021<br>development-era<br>manual-reference<br>analysis                                                                                          |
|-----------------|---------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Particle filter | Growth-rate evolution     | Random walk<br>( $\rho = 1$ )                                                          | Nonparametric<br>rate-versus-height<br>field from<br>2024–2025<br>pole-stream<br>increments; field<br>adopted before<br>growth noise is<br>added |
| Particle filter | Outlier probability $\pi$ | 0.0512                                                                                 | 0.065                                                                                                                                            |
| Particle filter | Outlier SD                | 70.62 cm                                                                               | 59.04 cm                                                                                                                                         |
| Particle filter | Resampling rule           | Systematic<br>resampling when<br>$ESS < 0.5M$                                          | Same                                                                                                                                             |
| Particle filter | Posterior interval        | Weighted 0.025<br>and 0.975<br>height-particle<br>quantiles after the<br>current image | Same                                                                                                                                             |

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*Table S3 continued*

| Component         | Quantity or operation | 2024–2025 daily<br>pole stream                                                                                                                                                                                                   | 2021<br>development-era<br>manual-reference<br>analysis                                                                                             |
|-------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Particle filter   | Predictive interval   | Numerical 0.025<br>and 0.975<br>quantiles of the<br>complete<br>particlewise<br>Gaussian mixture                                                                                                                                 | Same algorithm;<br>field-height<br>comparison<br>emphasizes the<br>posterior interval                                                               |
| Measurement<br>SD | Inlier scale          | Prespecified root-<br>sum-of-squares<br>budget: 8-cm<br>pose term, pole-fit<br>term, 1.5 cm per<br>band outside<br>fitted range, and<br>band-regularity<br>term; the total is<br>multiplied by 0.85<br>after depth<br>correction | $12[1 + 1.5g_t]$ cm,<br>where $g_t$ is the<br>observed gust<br>normalized over<br>the five evaluated<br>dates; range<br>12–30 cm, fallback<br>15 cm |

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Table S3 continued

| Component             | Quantity or operation | 2024–2025 daily<br>pole stream                                                                          | 2021<br>development-era<br>manual-reference<br>analysis                                                                                  |
|-----------------------|-----------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Measurement<br>SD     | Interpretation        | Pilot uncertainty<br>model; process<br>and measurement<br>variation are not<br>separately<br>identified | Retrospective<br>full-window<br>weather<br>normalization;<br>prospective use<br>requires<br>externally frozen<br>normalization<br>limits |
| Primary<br>comparison | Empirical sample      | Next<br>image-derived<br>measurements<br>after each first<br>observation                                | All 132 records,<br>33 plants, 12 rows,<br>and 5 dates                                                                                   |
| Primary<br>comparison | Cluster uncertainty   | Cameras for<br>calibration<br>summaries                                                                 | 20,000<br>row-bootstrap<br>replicates; seed<br>20260908<br>(endpoint seed<br>20260909)                                                   |

### 66 S3. Causal online definition

67 Let  $I_t$  be the image acquired at time  $t$ , and let  $\mathcal{I}_{\leq t} = \{I_1, \dots, I_t\}$  denote the accumulated  
68 image pool. With detector weights, thresholds, reference definitions, geometry, and filter

69 settings fixed, the operational update is

$$\mathcal{O}_t = A(I_t, \mathcal{O}_{t-1}; \theta_{\text{fixed}}), \quad \mathcal{O}_t \text{ is measurable with respect to } \mathcal{I}_{\leq t}. \quad (\text{S1})$$

70 Here  $\mathcal{O}_t$  contains the association state, physical image measurement, and height-growth  
 71 particle distribution returned immediately after  $I_t$  is added. An image  $I_{t+k}$ ,  $k > 0$ , is absent  
 72 from Equation S1. When it arrives, it initiates a new update; it does not revise  $\mathcal{O}_t$ .

The field association step predicts each track by carrying forward its running horizontal position  $m_{t-1}^x$  and base row  $m_{t-1}^b$ . Current detections are assigned to these carried states by minimum-cost matching. For an accepted observation  $(x_t, b_t)$ ,

$$m_t^x = (1 - K)m_{t-1}^x + Kx_t, \quad (\text{S2})$$

$$m_t^b = (1 - K)m_{t-1}^b + Kb_t, \quad K = 0.5. \quad (\text{S3})$$

73 Equation S3 stores means only. The code does not store a position covariance, so  $m_t^x$  and  $m_t^b$   
 74 are called running position states rather than position posteriors in this supplement.

75 The physical reference attached to a plant image is also selected as of  $t$ : only accepted  
 76 reference fits whose calibration time is no later than the plant-image capture time are eligible.  
 77 The selected top and base pixels are converted to centimeters, the depth factor is applied,  
 78 and the resulting measurement updates the height-growth particle distribution. The cached  
 79 detection-to-measurement audit recomputed 391 camera-time prefixes and compared 51,035  
 80 previously emitted rows, with 0 mismatches.

81 Primary manual-reference validation uses 2021 plants from 12 stationary-camera rows.  
 82 These subjects are absent from 2024–2025 development data, and no manual height enters the  
 83 image measurement, growth field, filter, noise parameters, or comparator settings. The five  
 84 2021 reference dates end with the season-end measurement and only score estimates. Newly  
 85 completed pole annotations describe 2021 images. Thus this is independent, label-held-out,  
 86 subject-disjoint validation. The separate 2024-to-2025 test assesses year-forward uncertainty;  
 87 every online update excludes future images.

88 In the simulation-only candidate extension, candidate  $j$  with detector evidence  $c_{tj}$  and

89 implied height  $y_{tj}$  is scored before the measurement update as

$$j_t^* = \arg \max_j \{c_{tj} p(y_{tj} \mid \mathcal{I}_{\leq t-1})\}. \quad (\text{S4})$$

90 This operation lets the height prior affect which top candidate becomes the measurement. It  
91 is distinct from applying the particle filter after an independently chosen detection, and it  
92 was not used to generate the empirical field-height table.

#### 93 **S4. Manual 2021 support-pole annotations**

94 A study annotator manually created the physical-reference annotations. The source  
95 annotation directory contains one CVAT XML file for each of 12 camera rows, named  
96 `C-[camera-row number].xml`. An XML can contain multiple dated images, and each image  
97 can contain several polylines labeled `pole1` through `pole4`. These labels distinguish visible  
98 camera-support poles within an image. The numeric suffix does not specify pole depth or  
99 physical length.

100 For each polyline, the importer reads all vertices, takes the smallest image row as the  
101 nominal camera-mount endpoint and the largest image row as ground contact, and records the  
102 sum of segment lengths in pixels. The physical definition is the nominal vertical distance from  
103 ground contact to the camera mounting or optical-center height: 5 ft (152.4 cm). Annotation  
104 status is read from the CVAT attribute named `status`. The imported data contain 199  
105 traces: 172 marked usable and 27 marked unusable.

106 An earlier export contained two filenames for the same C-039 scene. Following the project  
107 decision, `C-039_2021-07-30.JPG` was retained and `C-039_2021-07-30JPG.JPG` was removed.  
108 The updated `C-039.xml` also includes `LH54.21-C-039_2021-07-12-10-57_CAM827.JPG`. The  
109 curated source therefore contains 61 unique XML-referenced images, and every referenced  
110 image was found in the source image directory.

111 The installation records separately specify four horizontal field-layout intervals of 35 in,  
112 totaling 140 in between adjacent rows. These are not graduations on the vertical support  
113 poles. They also differ from the 1-ft red-band spacing on the 2024–2025 calibration poles.  
114 The 2021 import is consequently an installation and annotation resource; it is not pooled

with red-band calibration observations.

For the temporal repeatability audit, usable annotations were grouped by camera row and pole label. The 172 traces formed 39 series with at least two dated images across 12 rows. Within each series, the sample standard deviation of polyline length was divided by its mean. The median coefficient of variation was 1.24% (95% whole-row bootstrap interval, 0.78–1.99%), and the 90th percentile was 3.65%. 38 of 39 series were below 5%; the maximum was 11.51%. The calculation uses all traces marked usable and is descriptive of projected-length stability. It neither removes the largest series nor treats temporal repeatability as proof of the nominal 5-ft length.

The importer writes three canonical products:

- `data/processed/manual_poles_2021/manual_pole_annotations_2021.csv`, one row per retained trace;
- `data/processed/manual_poles_2021/pole_definitions_2021.csv`, the physical definitions and source records; and
- `data/processed/manual_poles_2021/summary.json`, counts, duplicate decision, and image-presence audit.
- `outputs/pole_repeatability_2021/`, per-series coefficients of variation and the whole-row bootstrap summary.

## S5. Physical calibration and depth correction

The 2024–2025 red-band pole supplies positions with known 30.48-cm spacing. Within one reference image, band image rows  $y_j$  are centered and scaled as  $u_j = (y_j - \bar{y})/s_y$ , and the calibration curve is

$$z(y) = \frac{au + b}{cu + 1}. \quad (\text{S5})$$

Only height differences are used. Consequently, an arbitrary additive physical datum along the pole does not affect a top-minus-base extent. Held-out bands evaluate interpolation, while the root-direction diagnostic removes lower bands, fits retained upper bands, and predicts downward. The reverse lower-to-upper calculation is not evidence about root extrapolation.

For the depth correction, place a camera at physical height  $H$ . Under the ideal pinhole model with negligible roll and a level shared ground plane, the image row of a point at height  $z$  and horizontal distance  $D$  can be written, up to the image-row sign convention, as

$$y(z, D) - y_{\text{hor}} = f_{\text{px}} \frac{H - z}{D}. \quad (\text{S6})$$

For two points separated by a vertical extent  $h = z_{\text{top}} - z_{\text{base}}$ , subtraction removes  $H$  and the horizon row:

$$|\Delta y(D)| = \frac{f_{\text{px}}}{D} h. \quad (\text{S7})$$

A scale learned at reference depth  $D_{\text{ref}}$  would assign a target pixel extent the value

$$h_{\text{pole scale}} = \frac{D_{\text{ref}}}{f_{\text{px}}} |\Delta y(D_{\text{tgt}})| = \frac{D_{\text{ref}}}{D_{\text{tgt}}} h_{\text{target}}. \quad (\text{S8})$$

Therefore,

$$h_{\text{target}} = \kappa h_{\text{pole scale}}, \quad \kappa = \frac{D_{\text{tgt}}}{D_{\text{ref}}}. \quad (\text{S9})$$

For the documented 2024–2025 geometry,  $D_{\text{tgt}} = 8.5$  ft and  $D_{\text{ref}} = 10.0$  ft, so  $\kappa = 0.85$ . The measurement standard deviation, being in the same linear units, is multiplied by the same factor. The factor is treated as fixed in the reported analysis; uncertainty in measured distances is not propagated.

Equation S9 is exact only within the model assumptions used in the derivation. Camera tilt or roll, lens distortion, uneven ground, different plant depths within a row, uncertain distances, and within-season camera movement can produce residual error.

## S6. Height-growth particle filter and exact predictive mixture

For elapsed time  $\Delta_t$  in days, the implemented transition order is

$$h_t^{(m)} = \text{clip}_{[0,450]} \left[ h_{t-1}^{(m)} + \Delta_t r_{t-1}^{(m)} + \sigma_h \sqrt{\Delta_t} \epsilon_{h,t}^{(m)} \right], \quad (\text{S10})$$

$$r_t^{(m)} = \text{clip}_{[-4,15]} \left[ T \left( h_t^{(m)}, r_{t-1}^{(m)} \right) + \sigma_r \sqrt{\Delta_t} \epsilon_{r,t}^{(m)} \right], \quad (\text{S11})$$

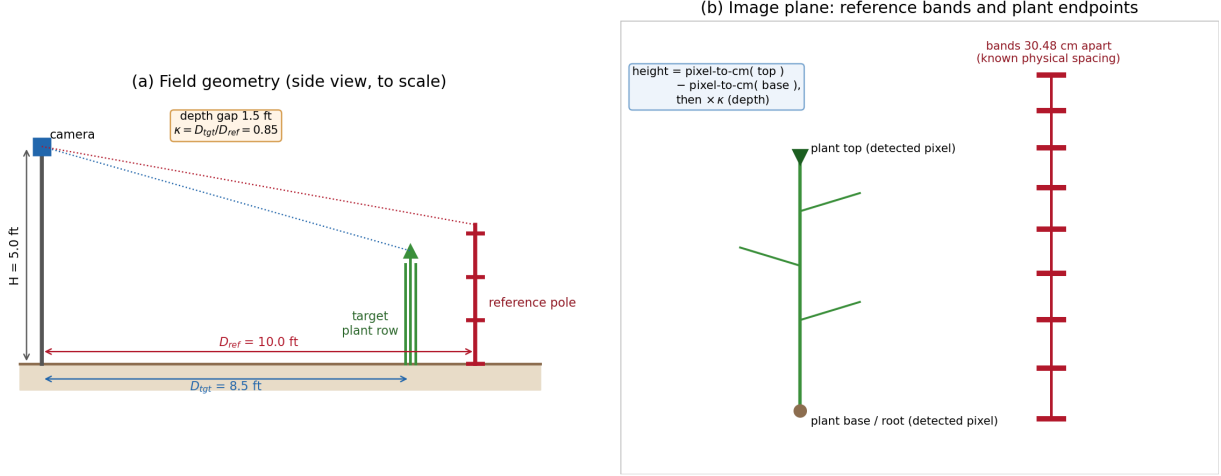


Figure S1: Idealized calibration-transfer geometry. (a) Side view with the 2024–2025 camera at 5 ft, the target row at  $D_{\text{tgt}} = 8.5$  ft, and the red-band reference at  $D_{\text{ref}} = 10.0$  ft, giving  $\kappa = 0.85$ . (b) Image-plane construction: the projective curve maps detected top and base rows onto the pole scale before the fixed depth factor is applied. The 30.48-cm labels refer to the red-band calibration pole, not the separate 2021 support poles.

where the innovations are independent standard normal variables. For the daily pole stream,  $T(h, r) = r$ . For the 2021 transferred-growth-field configuration,  $T(h, r) = \hat{r}(h)$ , the nonparametric rate-versus-height field estimated from 2024–2025 image increments. Manual 2021 field heights do not enter  $\hat{r}$ .

Conditional on height particle  $h_t^{(m)}$ , the observation law is

$$p(Y_t = y \mid h_t^{(m)}) = (1 - \pi_t) \phi(y; h_t^{(m)}, \sigma_t^2) + \pi_t \phi(y; h_t^{(m)}, \sigma_{\text{out}}^2). \quad (\text{S12})$$

Let  $w_m^-$  be normalized particle weights after propagation and before observing  $Y_t$ . The exact one-step predictive density and cumulative distribution are

$$p_t^-(y) = \sum_{m=1}^M w_m^- \left[ (1 - \pi_t) \phi(y; h_t^{(m)}, \sigma_t^2) + \pi_t \phi(y; h_t^{(m)}, \sigma_{\text{out}}^2) \right], \quad (\text{S13})$$

$$F_t^-(y) = \sum_{m=1}^M w_m^- \left[ (1 - \pi_t) \Phi\left(\frac{y - h_t^{(m)}}{\sigma_t}\right) + \pi_t \Phi\left(\frac{y - h_t^{(m)}}{\sigma_{\text{out}}}\right) \right]. \quad (\text{S14})$$

Predictive endpoints at probabilities 0.025, 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95, and 0.975 solve  $F_t^-(q_p) = p$  by numerical root finding. These quantiles provide central 50%, 80%, 90%, and 95% intervals. The predictive probability integral transform is  $F_t^-(Y_t)$ . The reported

164 predictive standard deviation uses

$$\text{Var}(Y_t \mid \mathcal{I}_{\leq t-1}) = \sum_m w_m^- \left( h_t^{(m)} - \bar{h}_t^- \right)^2 + (1 - \pi_t) \sigma_t^2 + \pi_t \sigma_{\text{out}}^2. \quad (\text{S15})$$

165 Equations S14 and S15 retain both mixture components. They replace the earlier normal  
166 approximation based only on latent-particle and inlier variances.

167 After  $Y_t$  is observed, the weights are updated as

$$w_m \propto w_m^- p(Y_t \mid h_t^{(m)}), \quad \sum_m w_m = 1. \quad (\text{S16})$$

168 The posterior mean and interval for latent height are computed from these updated weights.  
169 A posterior interval therefore targets  $h_t$  after the current image, whereas the predictive  
170 interval targets the next noisy image-derived measurement before it is observed. Coverage  
171 for those two targets is not interchangeable.

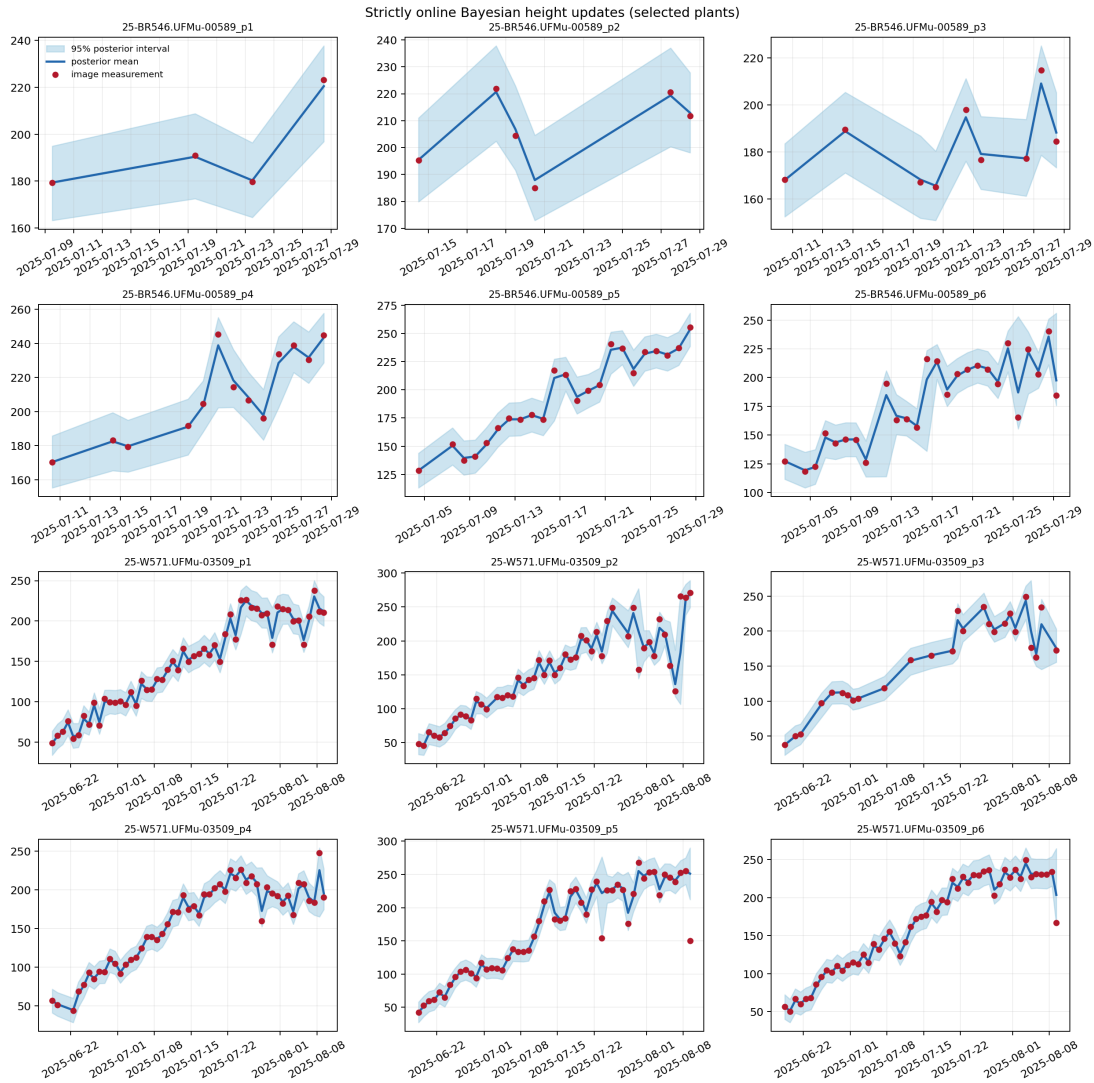


Figure S2: Selected 2025 daily pole-camera trajectories from the field-executed height-growth particle filter. Red points are depth-corrected image measurements, blue lines are posterior means after the corresponding image, and shading is the weighted 95% latent-height posterior interval. Later observations initiate later updates; they do not retrospectively revise the plotted earlier estimates. The figure illustrates posterior behavior against image-derived measurements and is not validation against manual plant height.

## S7. Uncertainty against manual height and across years

This secondary analysis evaluates temporal transfer of predictive uncertainty for the next image-derived measurement. The 0.85 depth conversion changes every quantity expressed in centimeters. The corrected analysis therefore applies it to the image-derived height and its measurement standard deviation, and estimates the process and outlier scales from already corrected 2024 increments. The 2025 records are not used to estimate these values. Because plants were newly grown each season, the 2025 cohort is temporally later and subject-disjoint from 2024; no biological plant subject appears in both years. Over the complete quality-controlled stream, the unit-consistent predictive mean had mean absolute error 15.13 cm, mean log score -4.440, and weighted interval score 8.08 cm.

Table S4 and Figure S3 show calibration at multiple levels. On 372 year-forward 2025 forecasts from 11 evaluated plant tracks and 2 cameras, the 95% interval achieved 94.9% coverage with a 97.8-cm mean width. The 50%, 80%, and 90% intervals covered 65.1%, 86.3%, and 92.7%, respectively. This pattern is conservative at the lower levels even though the 95% result is near nominal. Subject disjointness prevents repeated plants across years; it does not make repeated forecasts within a plant or camera independent. With only two test cameras, this analysis supports temporal transfer for those installations and does not establish broad camera generalization.

The same table adds the 2021 development-era physical-reference target. These intervals summarize latent height after the current image has been incorporated, and manual height is used only for evaluation. Central 80% intervals covered 87.1% of the 132 records with a 47.9-cm mean width; the 95% intervals covered 94.7% with a 75.4-cm mean width. Whole-camera-row bootstrap intervals were 79.3–93.6% and 89.4–98.6% for coverage, respectively. Per-date results and interval-width bootstrap summaries are released in `outputs/manual_height_uncertainty_2021/`.

## S8. Matched real-image candidate audit

The real-image audit uses cached candidates from the released 2021 pose checkpoint. Each temporal method receives identical candidates, confidence scores, association gates, fixed

Table S4: Multi-level uncertainty evaluation. The 2024–2025 rows evaluate one-step predictive intervals for the next image-derived measurement. The 2021 rows evaluate posterior intervals for latent height after the current image against held-out manual height.

| Evaluation            | Records | Clusters | Nominal (%) | Coverage (%) | Mean width (cm) |
|-----------------------|---------|----------|-------------|--------------|-----------------|
| 2024 development      | 1266    | 6        | 50          | 63.7         | 30.7            |
|                       |         |          | 80          | 87.8         | 59.4            |
|                       |         |          | 90          | 94.1         | 78.4            |
|                       |         |          | 95          | 96.8         | 98.2            |
| 2025 test             | 372     | 2        | 50          | 65.1         | 30.6            |
|                       |         |          | 80          | 86.3         | 59.3            |
|                       |         |          | 90          | 92.7         | 78.2            |
|                       |         |          | 95          | 94.9         | 97.8            |
| 2021 manual reference | 132     | 12 rows  | 80          | 87.1         | 47.9            |
|                       |         |          | 95          | 94.7         | 75.4            |

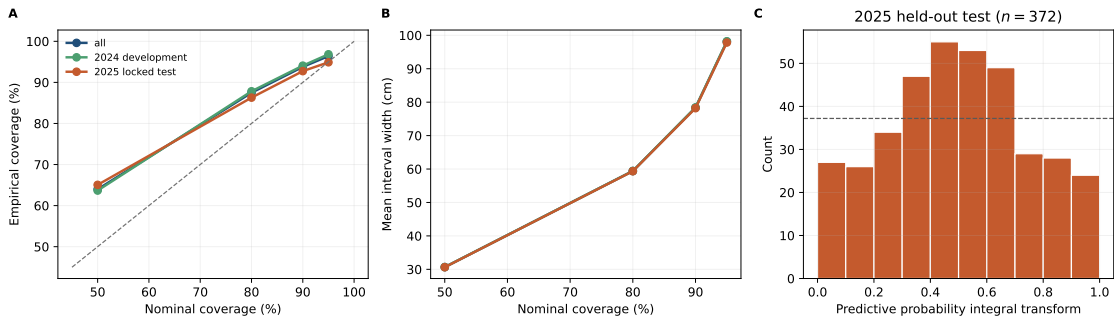


Figure S3: Secondary temporal evaluation of coverage and interval width for the unit-consistent predictive distribution. Parameters were estimated from the 2024 development cohort; the newly planted, subject-disjoint 2025 cohort was held out. The diagonal in the coverage panel marks exact nominal coverage. Points above the diagonal indicate conservative intervals.

scale (0.415886 cm/pixel), and particle-filter settings. Human-mask root positions initialize the first image only. Subsequent associations use the running state; manual field height is reserved for scoring. The matched subset contains 63 plant-dates from 17 tracks in 6 cameras.

The post-detection and prior-in-candidate methods agreed on every selected candidate (Table S5). Their errors are consequently identical. This is a useful negative control: under these natural-image candidates and gates, there was no ambiguity for the height distribution to resolve. It cannot verify the accuracy gain seen in the controlled ambiguity simulation.

Table S5: Real-image candidate audit. Manual field heights were used only for scoring.

| Method                                             | Signed error (cm) | MAE (cm) | RMSE (cm) |
|----------------------------------------------------|-------------------|----------|-----------|
| Single frame                                       | -9.55             | 13.38    | 18.01     |
| Identical candidate plus post-detection filter     | -9.27             | 13.07    | 17.72     |
| Height prior in candidate scoring plus same filter | -9.27             | 13.07    | 17.72     |

## S9. Field-error distribution and row influence

The primary comparison uses record-weighted mean absolute error and its whole-row bootstrap interval. As a secondary robustness audit, Table S6 reports a complete set of absolute-error quantiles for every fixed causal estimator. The robust Bayesian particle filter had the lowest error at each reported quantile. Its median absolute error was 7.30 cm, compared with 8.74 cm for the single-frame extent. The paired 1.43-cm median reduction had a 0.29 to 3.70 cm whole-row bootstrap interval.

Influence was evaluated by removing each stationary-camera row in turn and recomputing the record-weighted mean absolute-error difference. All 12 leave-one-row-out estimates favored the particle filter, ranging from 0.61 to 1.42 cm. These checks show that the direction of the observed advantage is not created by one camera row or only by extreme errors. They remain secondary to the prespecified mean absolute-error comparison, whose cluster interval includes zero.

Table S6: Secondary absolute-error distribution for the independent 2021 manual-reference validation. Values are centimeters across all 132 plant-date records; lower values are better.

| Estimator                       | 25th | Median | 75th  | 90th  | 95th  |
|---------------------------------|------|--------|-------|-------|-------|
| Single-frame image extent       | 3.35 | 8.74   | 15.32 | 26.01 | 31.51 |
| EWMA                            | 5.38 | 11.92  | 21.08 | 29.87 | 37.25 |
| Running median                  | 5.37 | 12.13  | 25.26 | 42.97 | 65.61 |
| Holt level-trend                | 5.24 | 9.85   | 20.34 | 28.57 | 35.30 |
| Gaussian local-linear           | 3.40 | 8.41   | 15.56 | 25.94 | 31.76 |
| Kalman                          |      |        |       |       |       |
| Robust Bayesian particle filter | 3.31 | 7.30   | 15.15 | 23.35 | 30.35 |

## S10. All-plant mature-date results

Table S7 lists the last available 2021 observation for every plant retained in the 2021 development-era manual-reference validation. No plant is removed using image-minus-field residuals. The single-frame column is the archived geometrically scaled mask extent; the Bayesian column is the height-growth particle-filter posterior mean. The interval is the 2.5th to 97.5th percentile of the latent-height posterior after the endpoint image. Manual endpoint height is used only to score the corresponding prediction and is not supplied to the filter.

Table S7: All 33 plants at their last available 2021 field date. Heights and interval endpoints are in centimeters.

| Plant    | Manual field | Single frame | Bayesian mean | Bayesian 95% interval |
|----------|--------------|--------------|---------------|-----------------------|
| C_004-p1 | 110.0        | 110.8        | 109.2         | [79.2, 142.4]         |
| C_004-p3 | 132.0        | 129.8        | 149.0         | [120.3, 178.9]        |
| C_004-p4 | 140.0        | 144.5        | 138.0         | [108.4, 169.2]        |
| C_005-p1 | 144.0        | 153.9        | 128.3         | [98.5, 158.7]         |
| C_005-p4 | 130.0        | 125.6        | 136.4         | [106.8, 167.3]        |
| C_005-p5 | 138.0        | 133.9        | 136.6         | [106.7, 167.7]        |
| C_009-p1 | 148.0        | 146.8        | 151.1         | [121.6, 182.2]        |
| C_009-p2 | 132.0        | 133.5        | 140.1         | [111.9, 170.8]        |
| C_009-p3 | 136.0        | 146.0        | 154.7         | [124.6, 185.4]        |
| C_010-p1 | 116.0        | 116.0        | 123.2         | [92.1, 157.1]         |
| C_010-p2 | 112.0        | 120.8        | 127.0         | [95.9, 161.0]         |
| C_010-p3 | 70.0         | 98.1         | 80.3          | [28.5, 136.8]         |
| C_011-p1 | 196.0        | 211.9        | 204.0         | [141.2, 239.5]        |
| C_011-p2 | 196.0        | 210.4        | 205.6         | [137.7, 244.5]        |
| C_011-p3 | 196.0        | 216.7        | 198.6         | [149.6, 231.2]        |
| C_021-p3 | 48.0         | 75.9         | 45.3          | [11.6, 79.1]          |
| C_021-p4 | 110.0        | 86.9         | 117.8         | [86.8, 159.0]         |

*Continued on next page*

Table S7 continued

| Plant    | Manual field | Single frame | Bayesian mean | Bayesian 95% interval |
|----------|--------------|--------------|---------------|-----------------------|
| C_024-p1 | 116.0        | 137.2        | 122.6         | [91.1, 157.4]         |
| C_024-p2 | 112.0        | 122.7        | 130.2         | [99.5, 164.1]         |
| C_024-p3 | 118.0        | 123.9        | 125.0         | [94.8, 158.8]         |
| C_029-p1 | 210.0        | 212.1        | 208.7         | [178.5, 238.0]        |
| C_029-p4 | 200.0        | 217.9        | 216.9         | [185.9, 246.8]        |
| C_029-p5 | 190.0        | 216.1        | 212.8         | [182.7, 242.6]        |
| C_032-p1 | 130.0        | 124.6        | 125.2         | [93.8, 158.6]         |
| C_032-p2 | 120.0        | 136.4        | 138.4         | [107.2, 170.6]        |
| C_032-p4 | 140.0        | 142.6        | 129.7         | [99.2, 164.6]         |
| C_037-p2 | 90.0         | 91.9         | 113.4         | [83.5, 145.5]         |
| C_037-p3 | 100.0        | 98.1         | 117.4         | [88.8, 148.8]         |
| C_037-p4 | 112.0        | 115.8        | 110.6         | [80.6, 144.4]         |
| C_038-p1 | 142.0        | 84.8         | 83.2          | [51.9, 116.8]         |
| C_039-p1 | 120.0        | 154.7        | 147.1         | [117.6, 176.6]        |
| C_039-p2 | 110.0        | 155.1        | 144.4         | [113.8, 176.6]        |
| C_039-p3 | 140.0        | 147.2        | 136.7         | [107.0, 167.5]        |

The all-record analysis includes first images so that every available manual reference remains in the primary estimand. A secondary analysis isolates the 99 records after a plant's first image, when longitudinal information from a preceding image exists. Across these records, single-frame and Bayesian mean absolute errors were 12.58 and 11.14 cm. The 1.44-cm reduction had a whole-row bootstrap interval of -0.21 to 3.26 cm. This outcome-independent restriction preserves the direction of the primary estimate but remains statistically imprecise.

Across these 33 endpoints, single-frame mean absolute error was 14.45 cm and the Bayesian mean absolute error was 12.38 cm. The Bayesian signed error was +6.17 cm. The stationary-camera-row bootstrap interval for the 2.08-cm within-plant mean-absolute-error reduction

was -1.47 to 6.82 cm. These endpoint results are secondary because the interval is wide and includes zero; the main article uses all 132 plant-date records as its primary comparison.

## S11. Data provenance and reproducibility

All paths below are relative to the anonymous review archive supplied with the submission.

Table S8: Data and result provenance for the revised submission.

| Asset                         | Canonical path or source                                                       | Role and limitation                                                                                                                                                                                                                                                                                             |
|-------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2021 pole XML                 | <code>data/manual_annotations/poles_2021/C-*.xml</code>                        | Human physical-reference annotations; one XML per camera row                                                                                                                                                                                                                                                    |
| 2021 pole images              | <code>data/raw/pole_calibration_images/</code>                                 | All 61 curated images; confirmed duplicate omitted                                                                                                                                                                                                                                                              |
| Imported 2021 pole table      | <code>data/processed/manual_poles_2021/manual_pole_annotations_2021.csv</code> | Trace endpoints, status, pixel length, and physical definition after duplicate removal                                                                                                                                                                                                                          |
| Pole temporal repeatability   | <code>outputs/pole_repeatability_2021/</code>                                  | All 39 repeated usable row-pole series; within-series pixel-length CV and whole-row bootstrap summary                                                                                                                                                                                                           |
| Pole definitions              | <code>data/processed/manual_poles_2021/pole_definitions_2021.csv</code>        | Documents 5-ft support length and distinguishes horizontal 35-in marks from red-band spacing                                                                                                                                                                                                                    |
| Daily image manifest          | <code>data/processed/pole_camera_daily_manifest.csv</code>                     | As-of capture records, local/remote identifiers, and checksums                                                                                                                                                                                                                                                  |
| Released pose checkpoint      | <code>models/maize_pose_2021_best.pt</code>                                    | 2021 YOLO pose weights with SHA-256 checksum; embedded software version, seed, deterministic setting, training arguments, and metrics are recorded in <code>models/checkpoint_metadata.json</code> , while the referenced dataset file, exact image split, original masks, and source commit remain unavailable |
| Curated-image pose candidates | <code>outputs/pose_candidates_annotation_set_2021/pose_candidates.csv</code>   | Complete cache for all 61 released images                                                                                                                                                                                                                                                                       |
| Automatic calibrations        | <code>data/processed/automatic_pole_calibrations_daily.csv</code>              | Candidate red-band fits and quality-control fields                                                                                                                                                                                                                                                              |
| Causal audit                  | <code>outputs/online_study/causality_audit_daily.json</code>                   | Prefix recomputation for cached detections through measurement assembly                                                                                                                                                                                                                                         |

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Table S8 continued

| Asset                                | Canonical path or source                                                                      | Role and limitation                                                                                                                                                                                         |
|--------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Daily Bayesian output                | outputs/online_study/pole_camera_bayesian_daily_revised/online_bayesian_height_posteriors.csv | Unit-consistent exact-mixture predictive and posterior summaries                                                                                                                                            |
| Daily model settings                 | outputs/online_study/pole_camera_bayesian_daily_revised/model_hyperparameters.json            | Complete saved particle-filter settings                                                                                                                                                                     |
| 2021 2021 development-era validation | data/raw/heights_compare_2021.csv                                                             | Archived image extents and 132 manual field measurements across 12 stationary-camera rows; 63 records across six rows have recovered source-frame coordinates, while 69 retain only the derived extent      |
| Partial source-frame recovery        | outputs/crop_positions_2021/crop_positions.csv                                                | Coordinates and masked-template match scores for 87 archived crop records across seven camera rows; 63 records join the development-era validation, and source pixels are not redistributed in this release |
| 2021 canonical posterior             | outputs/filter_height_sam_2021/filtered_height_sam__phi1.0_plus_pole_growth_field.csv         | Input to all-plants and per-plant summaries                                                                                                                                                                 |
| Primary field summary                | outputs/filter_height_sam_2021/all_plants_primary.json                                        | All-record and endpoint metrics with stationary-camera-row-cluster intervals; regenerate with <code>python reproduce_core.py</code>                                                                         |
| Manual-height interval audit         | outputs/manual_height_uncertainty_2021/                                                       | Central 80% and 95% posterior coverage, width, per-date summaries, and whole-row bootstrap intervals                                                                                                        |
| Detector-unit correction             | outputs/pixel_validation_2021/summary.json                                                    | Recomputed centimeter metrics using 0.41588595 cm/pixel rather than the withdrawn field-regression conversion                                                                                               |
| Field baselines                      | outputs/field_baselines_2021/                                                                 | Fixed causal comparators, absolute-error quantiles, row-influence audit, and row-cluster intervals                                                                                                          |
| Real-image candidate audit           | outputs/real_image_candidate_ablation_2021/                                                   | Matched candidate comparison with human truth used only for scoring                                                                                                                                         |